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Read all these books, documents and codes fully and carefully. As for the conversation I had with you earlier, I am interested in implementing the corrected equations of motion of Chapter 9 of the book with the corresponding tyre load equations too. However, I want to use https://github.com/zlijunting/TOTPT for such implementation. However, for TOTPT, the vehicle dynamics seem to have to be implemented with explicit dynamics. Can you have a look very carefully that whether the corrected equations of motion of Chapter 9 of the book can be implemented with TOTPT so that direct optimal control can apply to it?

已思考 3m 52s ›

Yes — **the corrected Chapter 9 EOM and tyre-load equations can be implemented in TOTPT for direct optimal control**, but **not as a drop-in replacement of the current** `tro_model_racecar.m`. The right conclusion is:

TOTPT's current structure is compatible with this model, but the Chapter 9 model should be implemented as an explicit state derivative plus algebraic/path constraints, using TOTPT's existing auxiliary-variable mechanism. This is the cleanest and most solver-friendly way.

TOTPT's repository describes the TRO part as a time-optimal raceline NLP using direct collocation, with MATLAB/CasADi and IPOPT in the offline trajectory-optimization layer. GitHub In the uploaded `tro_code.m`, the dynamics are built as a CasADi function

$$f_{\text{dyn}}(x, u, z, p_v) \to (dx, s_f),$$

and the collocation constraints call this function at the collocation points. tro_code Importantly, TOTPT already has auxiliary variables z, currently $\bar{a}_x, \bar{a}_y$, and enforces algebraic consistency through path constraints such as $\bar{a}_x = a_x, \bar{a}_y = a_y$. tro_model_racecar That is exactly the pattern you need for Chapter 9.

The corrected Chapter 9 equations are usable only after applying the errata. The errata corrects the angular-momentum signs in Eq. (9.31)/(9.32), and corrects the vertical-force equation in Eq. (9.44) so the second aerodynamic vertical term is Z_2^a, not another Z_1^a. SVDErrata The book's Chapter 9 EOM are already in an explicit/linearized form for roll and pitch, and the compact form Eq. (9.40) is especially useful for implementation. 978-94-017-8533-4

The key implementation point

The Chapter 9 body EOM themselves can be solved explicitly. Define

$$L = h - d.$$

Given tyre/aero resultants X, Y, N, and suspension moments

$$L_M = -k_\phi \phi - c_\phi \dot{\phi} + mgL\phi,$$

$$M_M = -k_\theta \theta - c_\theta \dot{\theta},$$

you can solve the corrected compact Chapter 9 EOM as

$$\ddot{\theta} = \frac{M_M - LX}{J_y},$$

$$\dot{u} = \frac{X}{m} + vr - L\ddot{\theta},$$

$$D = J_x J_z - J_{zx}^2,$$

$$A_\phi = L_M + LY,$$

$$A_r = N - LX\phi,$$

$$\ddot{\phi} = \frac{J_z A_\phi + J_{zx} A_r}{D},$$

$$\dot{r} = \frac{J_{zx} A_\phi + J_x A_r}{D},$$

$$\dot{v} = \frac{Y}{m} - ur + L\ddot{\phi}.$$

So **the chassis acceleration part is not the real obstacle**. It can be written explicitly.

The real coupling is the **tyre-load closure**. Chapter 9 load transfer uses

$$\Delta Z_1 t_1 = Y_1 q_1 + k_{\phi 1}\phi + c_{\phi 1}\dot{\phi},$$

$$\Delta Z_2 t_2 = Y_2 q_2 + k_{\phi 2}\phi + c_{\phi 2}\dot{\phi},$$

and

$$\Delta Z = -\frac{Xh + J_y\ddot{\theta}}{l}.$$

The book explicitly says that the tyre constitutive equations are still the Chapter 3 tyre equations, while roll/pitch mainly update the load transfers; it also notes that roll/pitch do not directly change the tyre constitutive equations. 978-94-017-8533-4

That means F_z depends on Y_i, but Y_i depends on tyre forces, and tyre forces depend on F_z. This creates an algebraic loop if you try to make everything a pure $\dot{x} = f(x, u)$ function.

Best TOTPT-compatible formulation

Use this as the state vector, preferably with u, v instead of V, β, because Chapter 9 defines the velocity at the VIP:

$$x = [u,\ v,\ r,\ \phi,\ \dot{\phi},\ \theta,\ \dot{\theta},\ n,\ \xi,\ \omega_{fl}, \omega_{fr}, \omega_{rl}, \omega_{rr}]^T.$$

Then use auxiliary variables such as

$$z = [a_x,\ a_y,\ \dot{r},\ \ddot{\phi},\ \ddot{\theta},\ F_{z,fl}, F_{z,fr}, F_{z,rl}, F_{z,rr}]^T.$$

Inside `tro_model_racecar_ch9.m`, compute tyre slips, tyre forces, $X, Y, N, Y_1, Y_2, \Delta X_1, \Delta X_2$, and the tyre loads using the auxiliary F_z variables. Then define the explicit time derivatives as

$$\dot{u} = a_x + vr - L\ddot{\theta},$$

$$\dot{v} = a_y - ur + L\ddot{\phi},$$

$$\dot{r} = \dot{r},$$

$$\dot{\phi} = \dot{\phi},$$

$$\ddot{\phi} = \ddot{\phi},$$

$$\dot{\theta} = \dot{\theta},$$

$$\ddot{\theta} = \ddot{\theta}.$$

For the track-frame states, with s as independent variable,

$$s_f = \frac{dt}{ds} = \frac{1 - n\kappa}{u\cos\xi - v\sin\xi},$$

$$\dot{n} = u\sin\xi + v\cos\xi,$$

$$\dot{\xi} = r - \kappa\frac{ds}{dt} = r - \frac{\kappa}{s_f}.$$

Then TOTPT still uses

$$\frac{dx}{ds} = s_f \frac{dx}{dt}.$$

This matches the current TOTPT pattern, where the present model computes $s_f = (1 - n\kappa)/(V\cos(\xi + \beta))$ and returns `dx = sf*xdot`. tro_model_racecar

The path constraints should then enforce the corrected Chapter 9 equations and the tyre-load equations:

$$ma_x - X = 0,$$

$$ma_y - Y = 0,$$

$$J_x\ddot{\phi} - J_{zx}\dot{r} - ma_y L - L_M = 0,$$

$$J_y\ddot{\theta} + ma_x L - M_M = 0,$$

$$J_z\dot{r} - J_{zx}\ddot{\phi} + ma_x L\phi - N = 0.$$

For tyre loads:

$$Z_{10} = \frac{mga_2}{l}, \qquad Z_{20} = \frac{mga_1}{l},$$

$$\Delta Z = -\frac{Xh + J_y\ddot{\theta}}{l},$$

$$\Delta Z_1 = \frac{Y_1 q_1 + k_{\phi 1}\phi + c_{\phi 1}\dot{\phi}}{t_1},$$

$$\Delta Z_2 = \frac{Y_2 q_2 + k_{\phi 2}\phi + c_{\phi 2}\dot{\phi}}{t_2}.$$

Then enforce

$$F_{z,fl} = \frac{1}{2}(Z_{10} - Z_1^a + \Delta Z) - \Delta Z_1,$$

$$F_{z,fr} = \frac{1}{2}(Z_{10} - Z_1^a + \Delta Z) + \Delta Z_1,$$

$$F_{z,rl} = \frac{1}{2}(Z_{20} - Z_2^a - \Delta Z) - \Delta Z_2,$$

$$F_{z,rr} = \frac{1}{2}\left(Z_{20} - Z_2^a - \Delta Z\right) + \Delta Z_2.$$

Also enforce

$$F_{z,ij} \geq \epsilon_F$$

because the errata-corrected tyre feasibility condition requires nonnegative vertical load, and the current TOTPT friction constraints divide by F_z^2. The errata explicitly corrects the Chapter 3 tyre feasibility condition to include $Z_{ij} \geq 0$. SVDErrata

What has to change in TOTPT

You mainly need to modify four places.

First, replace `tro_model_racecar.m` with a Chapter 9 model file. The current file has only 9 states: $V, \beta, \gamma, n, \xi, \omega_{fl}, \omega_{fr}, \omega_{rl}, \omega_{rr}$. tro_model_racecar The Chapter 9 version should add roll and pitch states and probably switch from V, β to u, v.

Second, update `veh_params.m`. The current parameter file only has planar parameters such as $m, I_z, l_f, l_r, h_c, I_w, r_w, w_t$, tyre coefficients, and input bounds. veh_params You need to add at least

$$J_x,\ J_y,\ J_z,\ J_{zx},\ h,\ d,\ q_1,\ q_2,$$

$$k_{\phi 1}, k_{\phi 2}, c_{\phi 1}, c_{\phi 2}, k_\theta, c_\theta,$$

and front/rear aero vertical-load distribution.

Third, update `tro_code.m` only structurally: increase `nx`, increase `nz`, update `z_min/z_max`, update initial guesses, and replace the old two auxiliary acceleration constraints with the full Chapter 9 residuals. The collocation core can stay the same, because it already treats X_k, U_k, Z_k, X_{kj} as NLP variables and puts path constraints into `g`. tro_code

Fourth, update plotting/post-processing. The current plotting assumes exactly 9 states and two auxiliary variables $\bar{a}_x, \bar{a}_y$. tro_plot_results After adding roll/pitch and more z variables, the plotting indices must be changed.

My recommendation

For your purpose, I would **not** try to force the corrected Chapter 9 model into a pure ODE with no auxiliary variables. It is mathematically possible, but it would either require simplifying the tyre-load closure or embedding a nonlinear solve inside the CasADi dynamics evaluation. That is a bad tradeoff for IPOPT.

The best implementation is:

$$\text{explicit } dx/ds = f(x,u,z,\kappa) \quad + \quad \text{algebraic/path constraints } h(x,u,z,\kappa) = 0.$$

This is fully compatible with TOTPT's direct-collocation NLP style, and it preserves the corrected Chapter 9 tyre-load equations much more cleanly than trying to hide the load-transfer algebra inside `f_dyn`.

So the final answer is:

Yes, the corrected Chapter 9 model can be implemented in TOTPT and used for direct optimal control. But the correct implementation is a semi-explicit direct-collocation formulation using TOTPT's auxiliary variables and equality path constraints. It is not a simple drop-in explicit replacement of the current planar 9-state `tro_model_racecar.m`.

資料來源

Please carefully double check your recommendations. Moreover, if the above answer is not the best and if there are better formulation, please make sure: 1. Do not compromise the fidelity of the correct Chapter 9 EOM and corresponding tyre load equations; 2. Come up with the best formulation that would be most friendly to IPOPT, that is to say, introduce none or minimum nonlinearity or nonconvexity; 3. Make sure TOTPT can handle your formulation and direct optimal control of TOTPT can apply to it.

已思考 2m 41s ›

I double-checked it carefully, and I would **revise my previous recommendation**.

My earlier formulation with

$$z = [a_x, a_y, \dot{r}, \ddot{\phi}, \ddot{\theta}, F_{z,fl}, F_{z,fr}, F_{z,rl}, F_{z,rr}]$$

is **compatible with TOTPT**, but it is **not the best IPOPT-friendly formulation**. It introduces too many auxiliary variables and too many equality constraints. The better formulation is:

$$\text{explicit Chapter 9 differential dynamics} \quad + \quad \text{only tyre vertical loads } F_z \text{ as algebraic/lifted variables.}$$

This keeps the corrected Chapter 9 EOM fidelity, minimizes unnecessary variables, and fits TOTPT's direct-collocation structure.

1. What must be preserved from Chapter 9

The corrected Chapter 9 EOM are already almost explicit. The book gives the linearized roll/pitch/yaw vehicle equations in Eq. (9.39), and then rewrites the rotational equations compactly in Eq. (9.40). The key corrected structure is

$$m[(\dot{u} - vr) + H\ddot{\theta}] = X,$$

$$m[(\dot{v} + ur) - H\ddot{\phi}] = Y,$$

$$J_x\ddot{\phi} - J_{zx}\dot{r} - ma_yH = L_M,$$

$$J_y\ddot{\theta} + ma_xH = M_M,$$

$$J_z\dot{r} - J_{zx}\ddot{\phi} + ma_xH\phi = N,$$

where

$$H = h - d.$$

The book explicitly notes that the model is an explicit linearized equilibrium form for a vehicle that can roll and pitch, and that the compact Eq. (9.40) is equivalent under the same assumptions. 978-94-017-8533-4

The load-transfer equations must also be preserved. Chapter 9 states that load transfers now depend explicitly on roll/pitch angular accelerations, while the tyre constitutive equations themselves remain the Chapter 3 tyre equations. 978-94-017-8533-4 The Chapter 9 load-transfer equations are

$$\Delta Z_1 t_1 = Y_1 q_1 + k_{\phi 1}\phi + c_{\phi 1}\dot{\phi},$$

$$\Delta Z_2 t_2 = Y_2 q_2 + k_{\phi 2}\phi + c_{\phi 2}\dot{\phi},$$

and

$$\Delta Z = \frac{-Xh + J_y\ddot{\theta}}{l}.$$

The errata must be applied: Eq. (9.31)/(9.32) need angular-momentum sign corrections, and Eq. (9.44) must use $Z_1^a + Z_2^a$, not repeated Z_1^a. SVDErrata SVDErrata

2. The best formulation is not "acceleration-lifted"; it is "load-lifted"

The only unavoidable algebraic loop is this:

$$F_z \to F_x, F_y \to X, Y, N, Y_1, Y_2 \to \Delta Z, \Delta Z_1, \Delta Z_2 \to F_z.$$

So the minimum useful lifting is

$$z = [F_{z,fl}, F_{z,fr}, F_{z,rl}, F_{z,rr}]^T.$$

Do **not** lift $a_x, a_y, \dot{r}, \ddot{\phi}, \ddot{\theta}$ unless you are debugging. They can be computed explicitly from the corrected Chapter 9 EOM once X, Y, N are known.

Given tyre forces and moment resultants, define

$$L_M = -k_\phi\phi - c_\phi\dot{\phi} + mgH\phi,$$

$$M_M = -k_\theta\theta - c_\theta\dot{\theta}.$$

Then compute

$$a_x = \frac{X}{m}, \qquad a_y = \frac{Y}{m},$$

$$\ddot{\theta} = \frac{M_M - HX}{J_y}.$$

For roll/yaw, solve only this fixed 2×2 linear system:

$$\begin{bmatrix} J_x & -J_{zx} \\ -J_{zx} & J_z \end{bmatrix} \begin{bmatrix} \ddot{\phi} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} L_M + HY \\ N - HX\phi \end{bmatrix}.$$

Equivalently, with

$$D = J_xJ_z - J_{zx}^2,$$

$$A_\phi = L_M + HY, \qquad A_r = N - HX\phi,$$

$$\ddot{\phi} = \frac{J_zA_\phi + J_{zx}A_r}{D},$$

$$\dot{r} = \frac{J_{zx}A_\phi + J_xA_r}{D}.$$

Then

$$\dot{u} = a_x + vr - H\ddot{\theta},$$

$$\dot{v} = a_y - ur + H\ddot{\phi}.$$

So the differential dynamics remain explicit.

3. Recommended state vector

Use u, v, not V, β, because Chapter 9 defines the vehicle velocity at the VIP M. A good state vector is

$$x = [u,\ v,\ r,\ \phi,\ \dot{\phi},\ \theta,\ \dot{\theta},\ n,\ \xi,\ \omega_{fl}, \omega_{fr}, \omega_{rl}, \omega_{rr}]^T.$$

This is better than keeping TOTPT's current

$$[V, \beta, \gamma, n, \xi, \omega_{fl}, \omega_{fr}, \omega_{rl}, \omega_{rr}],$$

because the current model is a planar 9-state model and computes $v_x = V\cos\beta, v_y = V\sin\beta$. tro_model_racecar Chapter 9's variables are naturally u, v, r, ϕ, θ.

For the track-coordinate part, use

$$s_f = \frac{dt}{ds} = \frac{1 - n\kappa}{u\cos\xi - v\sin\xi},$$

$$\dot{n} = u\sin\xi + v\cos\xi,$$

$$\dot{\xi} = r - \frac{\kappa}{s_f}.$$

Then TOTPT still uses

$$\frac{dx}{ds} = s_f \dot{x}.$$

This matches the current TOTPT idea, where the existing model computes s_f and then returns `dx = sf*xdot`. tro_model_racecar

4. Best tyre-load closure

Use $z = F_z$ inside the tyre model. Compute tyre forces from slip and F_z, then compute

$$X = X_1 + X_2 - X_a,$$

$$Y = Y_1 + Y_2,$$

$$N = Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2.$$

Then predict the Chapter 9 vertical loads:

$$Z_1^0 = \frac{mga_2}{l}, \qquad Z_2^0 = \frac{mga_1}{l},$$

$$\Delta Z = \frac{-Xh + J_y\ddot{\theta}}{l},$$

$$Z_1 = Z_1^0 + \Delta Z - Z_1^a,$$

$$Z_2 = Z_2^0 - \Delta Z - Z_2^a,$$

$$\Delta Z_1 = \frac{Y_1 q_1 + k_{\phi 1}\phi + c_{\phi 1}\dot{\phi}}{t_1},$$

$$\Delta Z_2 = \frac{Y_2 q_2 + k_{\phi 2}\phi + c_{\phi 2}\dot{\phi}}{t_2}.$$

Then enforce only four algebraic equalities:

$$F_{z,fl} - \left(\frac{Z_1}{2} - \Delta Z_1\right) = 0,$$

$$F_{z,fr} - \left(\frac{Z_1}{2} + \Delta Z_1\right) = 0,$$

$$F_{z,rl} - \left(\frac{Z_2}{2} - \Delta Z_2\right) = 0,$$

$$F_{z,rr} - \left(\frac{Z_2}{2} + \Delta Z_2\right) = 0.$$

This is the minimum algebraic closure that preserves the Chapter 9 tyre-load equations.

5. Important improvement over the previous answer

TOTPT already has auxiliary variables Z_k, path constraints, and an NLP variable vector containing states, inputs, auxiliary variables, and collocation helper states. tro_code tro_code So this formulation is compatible.

But the current code uses the same $Z_k(:, k+1)$ inside the collocation dynamics for all collocation points in one interval. tro_code That is acceptable for the original simple model, but **not ideal for the Chapter 9 load closure**.

For the best implementation, add collocation algebraic variables too:

$$Z_{kj} \in \mathbb{R}^{4\times Nd}.$$

Then evaluate

$$f_{\mathrm{dyn}}(X_{kj}, U_k, Z_{kj}, \kappa_{kj})$$

at collocation points, and enforce the four load equations at both knot and collocation points. This is a standard direct-collocation DAE treatment. It does not compromise the Chapter 9 equations, and IPOPT can handle it because it is just more sparse NLP variables and sparse equality constraints.

So the best TOTPT structure is:

$$X_k,\ U_k,\ Z_k,\ X_{kj},\ Z_{kj}$$

not only

$$X_k,\ U_k,\ Z_k,\ X_{kj}.$$

6. Even more explicit option for current TOTPT tyre model

For the **current TOTPT tyre model**, there is an even cleaner possibility.

The uploaded tyre force functions have the structure

$$F_x = (d_{1x}F_z + d_{2x})\sin(C_x \arctan(B_x\lambda)),$$

$$F_y = -(d_{1y}F_z + d_{2y})\sin(C_y \arctan(B_y\alpha)).$$

So for fixed state/input/slip, tyre forces are **affine in F_z**. tro_model_racecar Therefore the four Chapter 9 load equations become a 4×4 linear system:

$$(I - A(x,u))F_z = b(x,u).$$

You can solve this explicitly inside CasADi:

$$F_z = (I - A)^{-1}b.$$

That gives a pure explicit model

$$\dot{x} = f(x,u)$$

with no z variables at all.

However, I would still prefer the lifted F_z-only formulation for your serious implementation, because it is more robust if you later replace the simple TOTPT tyre law with a more realistic load-dependent Magic Formula. The explicit 4×4 solve is excellent for the current simple tyre model, but less general.

7. IPOPT-friendly details

Use the friction constraint without dividing by F_z^2. The current TOTPT form is rational:

$$1 - \left(\frac{F_x^2}{\mu_x^2 F_z^2} + \frac{F_y^2}{\mu_y^2 F_z^2}\right) \geq 0.$$

A more IPOPT-friendly equivalent, assuming $F_z \geq 0$, is

$$\frac{F_x^2}{\mu_x^2} + \frac{F_y^2}{\mu_y^2} \leq F_z^2.$$

Also enforce

$$F_z \geq F_{z,\min} > 0$$

because the errata-corrected Chapter 3 tyre feasibility condition requires nonnegative vertical load, and the original friction workload calculation divides by F_z. SVDErrata

Final recommendation

The best formulation is:

$$\text{explicit corrected Chapter 9 EOM} + \text{four lifted tyre-load variables } F_z + \text{four Chapter 9 tyre-load equality constraints.}$$

For full collocation fidelity in TOTPT, add Z_{kj} at collocation points and enforce the load equations there too.

I would **not** use the acceleration-lifted formulation as the final implementation. It works, but it is not the cleanest. The cleanest and most IPOPT-friendly formulation is:

$$x = [u, v, r, \phi, \dot{\phi}, \theta, \dot{\theta}, n, \xi, \omega_{fl}, \omega_{fr}, \omega_{rl}, \omega_{rr}], \quad z = [F_{z,fl}, F_{z,fr}, F_{z,rl}, F_{z,rr}].$$

This preserves the corrected Chapter 9 EOM and tyre-load equations, introduces only the algebraic variables needed to break the unavoidable tyre-load loop, and remains fully compatible with TOTPT's direct-collocation/IPOPT framework.

資料來源

Please double check your suggested formulation, make sure again: 1. Do not compromise the fidelity of the correct Chapter 9 EOM and corresponding tyre load equations; 2. Come up with the best formulation that would be most friendly to IPOPT, that is to say, introduce none or minimum nonlinearity or nonconvexity; 3. Make sure TOTPT can handle your formulation and direct optimal control of TOTPT can apply to it. 4. Double check your suggested fully collocation fidelity in TOTPT that adding Zkj at collocation points and enforce the load equations there too. Is TOTPT currently doing this or do you suggest it yourself and why.

已思考 2m 39s ›

After double-checking the equations and the uploaded TOTPT code, my final recommendation is:

$$\textbf{Use a semi-explicit collocation DAE formulation: explicit Chapter 9 dynamics + lifted } F_z \textbf{ variables.}$$

But I need to correct and sharpen one point from my previous answer:

$$\textbf{TOTPT is not currently using } Z_{kj} \textbf{ at collocation points. I suggest adding it.}$$

TOTPT currently has knot-point auxiliary variables `Zk`, but no collocation-point auxiliary variables. Its collocation dynamics use `Xkj` at the collocation points while passing the same `Zk(:,k+1)` into all collocation evaluations in that interval. It also applies `h_eq` only at the knot points. tro_code So adding Z_{kj} is my recommended modification, not something TOTPT already does.

1. The corrected Chapter 9 fidelity requirement

The corrected Chapter 9 model should be treated as the book's small-roll/small-pitch, flat-road, rigid-tyre handling model. The errata must be applied: Eq. (9.31)/(9.32) require angular-momentum sign corrections, and Eq. (9.44) must use $Z_1^a + Z_2^a$, not repeated Z_1^a. SVDErrata SVDErrata

Use

$$H = h - d.$$

The corrected Chapter 9 acceleration definitions are

$$a_x = (\dot{u} - vr) + H\ddot{\theta},$$

$$a_y = (\dot{v} + ur) - H\ddot{\phi}.$$

The book's corrected compact rotational form is

$$J_x\ddot{\phi} - J_{zx}\dot{r} - ma_yH = L_M,$$

$$J_y\ddot{\theta} + ma_xH = M_M,$$

$$J_z\dot{r} - J_{zx}\ddot{\phi} + ma_xH\phi = N.$$

These come directly from Eq. (9.39)/(9.40). 978-94-017-8533-4

The Chapter 9 load-transfer equations must also be kept:

$$\Delta Z_1 t_1 = Y_1 q_1 + k_{\phi 1}\phi + c_{\phi 1}\dot{\phi},$$

$$\Delta Z_2 t_2 = Y_2 q_2 + k_{\phi 2}\phi + c_{\phi 2}\dot{\phi},$$

$$\Delta Z = \frac{-Xh + J_y\ddot{\theta}}{l}.$$

The book explicitly states that adding roll and pitch changes the load transfers, while the tyre constitutive equations themselves are not directly changed by roll/pitch. 978-94-017-8533-4

So the formulation must preserve these equations exactly.

2. Best state and auxiliary-variable choice

Use Chapter 9's natural velocity variables, not TOTPT's current V, β variables. The current TOTPT model uses

$$x = [V, \beta, \gamma, n, \xi, \omega_{fl}, \omega_{fr}, \omega_{rl}, \omega_{rr}]^T,$$

with only two auxiliary variables $\bar{a}_x, \bar{a}_y$. tro_model_racecar

For the Chapter 9 model, use

$$x = [u,\ v,\ r,\ \phi,\ \dot{\phi},\ \theta,\ \dot{\theta},\ n,\ \xi,\ \omega_{fl}, \omega_{fr}, \omega_{rl}, \omega_{rr}]^T.$$

Use only the four vertical tyre loads as algebraic variables:

$$z = [F_{z,fl}, F_{z,fr}, F_{z,rl}, F_{z,rr}]^T.$$

Do **not** lift $a_x, a_y, \dot{r}, \ddot{\phi}, \ddot{\theta}$ unless debugging. Those accelerations can be computed explicitly once X, Y, N are known.

3. Explicit Chapter 9 dynamics once F_z is known

At each knot or collocation point, compute tyre slips, tyre forces, and then

$$X, \qquad Y, \qquad N, \qquad Y_1, \qquad Y_2.$$

Then

$$a_x = \frac{X}{m}, \qquad a_y = \frac{Y}{m}.$$

Pitch acceleration is explicit:

$$\ddot{\theta} = \frac{M_M - HX}{J_y}.$$

Roll/yaw are obtained from a fixed 2×2 linear system:

$$\begin{bmatrix} J_x & -J_{zx} \\ -J_{zx} & J_z \end{bmatrix} \begin{bmatrix} \ddot{\phi} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} L_M + HY \\ N - HX\phi \end{bmatrix}.$$

With

$$D = J_xJ_z - J_{zx}^2,$$

$$A_\phi = L_M + HY, \qquad A_r = N - HX\phi,$$

the explicit solution is

$$\ddot{\phi} = \frac{J_zA_\phi + J_{zx}A_r}{D},$$

$$\dot{r} = \frac{J_{zx}A_\phi + J_xA_r}{D}.$$

Then the translational derivatives are

$$\dot{u} = \frac{X}{m} + vr - H\ddot{\theta},$$

$$\dot{v} = \frac{Y}{m} - ur + H\ddot{\phi}.$$

So the differential dynamics remain explicit:

$$\dot{x} = f(x, u, z).$$

This satisfies TOTPT's requirement that the dynamics function return explicit derivatives.

4. Tyre-load closure: the minimum-fidelity-preserving algebraic part

Use the lifted F_z variables in the tyre model. Then compute the Chapter 9 load equations and enforce four residuals:

$$Z_1^0 = \frac{mga_2}{l}, \qquad Z_2^0 = \frac{mga_1}{l},$$

$$\Delta Z = \frac{-Xh + J_y\ddot{\theta}}{l},$$

$$Z_1 = Z_1^0 + \Delta Z - Z_1^a,$$

$$Z_2 = Z_2^0 - \Delta Z - Z_2^a,$$

$$\Delta Z_1 = \frac{Y_1q_1 + k_{\phi 1}\phi + c_{\phi 1}\dot{\phi}}{t_1},$$

$$\Delta Z_2 = \frac{Y_2q_2 + k_{\phi 2}\phi + c_{\phi 2}\dot{\phi}}{t_2}.$$

Then enforce

$$F_{z,fl} - \left(\frac{Z_1}{2} - \Delta Z_1\right) = 0,$$

$$F_{z,fr} - \left(\frac{Z_1}{2} + \Delta Z_1\right) = 0,$$

$$F_{z,rl} - \left(\frac{Z_2}{2} - \Delta Z_2\right) = 0,$$

$$F_{z,rr} - \left(\frac{Z_2}{2} + \Delta Z_2\right) = 0.$$

This is the smallest algebraic closure that keeps the corrected Chapter 9 tyre-load equations.

5. IPOPT-friendliness

The best IPOPT-friendly formulation is **not** to solve the F_z loop inside `f_dyn`. It is better to lift F_z and enforce the four load equations as sparse algebraic constraints.

Why?

Solving the load loop explicitly would hide the algebraic coupling inside a dense symbolic inverse. With the current TOTPT tyre law, tyre forces are affine in F_z, so an explicit 4×4 linear solve is theoretically possible because the uploaded tyre model uses

$$F_x = (d_{1x}F_z + d_{2x})\sin(C_x \arctan(B_x\lambda)),$$

$$F_y = -(d_{1y}F_z + d_{2y})\sin(C_y \arctan(B_y\alpha)).$$

tro_model_racecar

But that explicit solve would create rational expressions and denser derivatives. The lifted formulation gives IPOPT more variables, but the constraints remain local and sparse. IPOPT usually handles that better.

Also replace the current friction constraint form

$$1 - \left(\frac{F_x^2}{\mu_x^2F_z^2} + \frac{F_y^2}{\mu_y^2F_z^2}\right) \geq 0$$

with the equivalent positive-load form

$$\frac{F_x^2}{\mu_x^2} + \frac{F_y^2}{\mu_y^2} - F_z^2 \leq 0, \qquad F_z \geq F_{z,\min} > 0.$$

The current TOTPT code divides by F_z^2, which is less friendly numerically. tro_code

6. The Z_{kj} issue: current TOTPT vs my recommendation

TOTPT currently defines

$$X_k, \quad U_k, \quad Z_k, \quad X_{kj},$$

but not Z_{kj}. The decision vector is currently

$$w = [X_k;\ U_k;\ Z_k;\ X_{kj}],$$

and the collocation dynamics are evaluated as

MATLAB

So the collocation states X_{kj} are different at each collocation point, but the auxiliary variables $Z_k(:, k+1)$ are reused for all collocation points inside the interval. Path constraints are then evaluated only at the knot grid through `h_eq(Xk,Uk,Zk,pv)`. tro_code

For the original TOTPT model, that is acceptable because $z = [\bar{a}_x, \bar{a}_y]$ is mainly used as an auxiliary acceleration consistency device. But for Chapter 9, $z = F_z$ is not just a smoothing/output variable. It is part of the physical tyre-force closure. Therefore:

$$\text{If } f_{\text{dyn}} \text{ is evaluated at } X_{kj}, \text{ then } F_z \text{ must also exist and be constrained at } X_{kj}.$$

So I recommend changing the NLP variables to

$$w = [X_k;\ U_k;\ Z_k;\ X_{kj};\ Z_{kj}].$$

Then use

$$f_{\mathrm{dyn}}(X_{kj}, U_k, Z_{kj}, \kappa_{kj})$$

inside the collocation loop, and enforce the Chapter 9 load equations at both knots and collocation points:

$$h_{\mathrm{load}}(X_k, U_k, Z_k) = 0,$$

$$h_{\mathrm{load}}(X_{kj}, U_k, Z_{kj}) = 0.$$

This is my suggestion, not current TOTPT behavior.

It is necessary if you want full collocation fidelity. Otherwise, the ODE residual would be enforced at collocation states using tyre loads that are only constrained at the knot. That would be a lower-fidelity approximation of the Chapter 9 algebraic load closure.

Final checked recommendation

Use this formulation:

$$x = [u, v, r, \phi, \dot{\phi}, \theta, \dot{\theta}, n, \xi, \omega_{fl}, \omega_{fr}, \omega_{rl}, \omega_{rr}]^T,$$

$$z = [F_{z,fl}, F_{z,fr}, F_{z,rl}, F_{z,rr}]^T.$$

Use explicit corrected Chapter 9 dynamics:

$$\dot{x} = f(x, u, z),$$

and enforce only the four corrected Chapter 9 tyre-load residuals:

$$h_{\mathrm{load}}(x, u, z) = 0.$$

For TOTPT, modify the collocation transcription to include

$$Z_{kj}$$

and enforce $h_{\mathrm{load}} = 0$ at collocation points too.

This is the best balance I see:

1. It preserves the corrected Chapter 9 EOM and tyre-load equations.
2. It avoids unnecessary acceleration-lifting.
3. It avoids hiding the load loop inside a dense symbolic solve.
4. It is compatible with TOTPT's CasADi/IPOPT NLP structure.
5. It requires a real modification to TOTPT: adding Z_{kj}, because TOTPT currently does **not** do that.